Unit II

2. Authentication of Electronic records & E-Governance

2.1 Authentication of Electronic records
2.2 Digital Signature
2.3 RSA Digital Signature
2.4 Hash Function
2.5 Working of Digital Signature
2.6 Electronic Governance

2.1 Authentication of Electronic records:

Authentication of electronic records refers to the process of **verifying the identity of the sender** and ensuring that the electronic document is **genuine, unaltered, and trustworthy**. Under the **Information Technology (IT) Act, 2000**, electronic records gain legal validity when properly authenticated.

Key Points:

1. **Use of Digital Signatures:**
   The IT Act recognizes **digital signatures** as the primary method to authenticate electronic records. A digital signature is created using the sender's **private key** and can be verified using the corresponding **public key**, ensuring authenticity.

2. **Ensures Integrity of Data:**
   Authentication confirms that the electronic record has **not been altered or tampered with** during transmission. Even a small change will invalidate the digital signature.

3. **Legal Recognition:**
   Section 3 of the IT Act legally recognizes digital signatures, making digitally signed electronic records valid and acceptable in courts, government processes, and business transactions.

4. **Use of Certifying Authorities (CAs):**
   Digital signatures are issued by licensed **Certifying Authorities** who verify the identity of the user. This adds trust to the authentication process.

5. **Applications:**
   Authentication of electronic records is used in **e-government services, e-commerce, online banking, tax filing, digital contracts**, and various secure communication systems.

In short:

Authentication of electronic records ensures that the document is **genuine, secure, legally valid, and unaltered**, mainly through the use of **digital signatures and public-key cryptography**.

2.2 Digital Signature:

A **digital signature** is an electronic method used to **authenticate the identity** of the sender and to ensure the **integrity** of an electronic document. It is created using **public key cryptography**, mainly through the use of a **private key** and verified using a **public key**. The **Information Technology (IT) Act, 2000** gives legal recognition to digital signatures in India.

Key Points:

1. **Authentication:**
   A digital signature verifies the identity of the sender. Since it is created using the sender's private key, only the owner can generate it.

2. **Integrity:**
   Digital signatures ensure that the content of an electronic record has **not been altered or tampered with** after signing. Any modification invalidates the signature.

3. **Non-Repudiation:**
   The sender cannot deny having signed the document because the signature is uniquely linked with their private key.

4. **How It Works:**
   The sender's computer generates a **hash** of the message.
   This hash is encrypted using the sender's **private key** to create the digital signature.
   The receiver uses the sender's **public key** to verify the signature.

5. **Legal and Practical Use:**
   Digital signatures are widely used in **e-filing of taxes**, **online banking**, **e-commerce**, **government services**, **signing PDF documents**, and **secure email communication**.
   Certifying Authorities (CAs) issue Digital Signature Certificates (DSCs) to users.

In short:
A digital signature is a **secure, legally recognized electronic signature** that provides **authentication, integrity, and non-repudiation** using public-key cryptography.

2.3 RSA Digital Signature:

The **RSA Digital Signature** is a technique that uses the **RSA public-key algorithm** to authenticate electronic documents. It ensures **authenticity, integrity, and non-repudiation** by using a pair of mathematically related keys: a **private key** for signing and a **public key** for verification.

Key Points:

1. **Uses Public-Key Cryptography:**
   RSA digital signatures are created using the sender's **private key**. The corresponding **public key** is used by the receiver to verify the signature. This ensures that only the genuine owner could have signed the document.

2. **Process of Creating a Digital Signature:**
   A **hash** (message digest) of the original message is generated using a hash function like SHA-256.
   This hash is then **encrypted with the sender's private key** to create the RSA digital signature.
   The signature is sent along with the message.

3. **Verification Process:**
   The receiver decrypts the signature using the sender's **public key** to recover the hash.
   The receiver also generates a fresh hash from the received message.
   If both hashes match, the signature is valid and the message is unaltered.

4. **Security Features:**

RSA signature provides:

Authentication (confirms the identity of the signer)

Integrity (detects any modification of data)

Non-repudiation (signer cannot deny signing)

Security depends on large prime factorization, making RSA computationally strong.

5. **Applications:**

Widely used in **digital certificates (SSL/TLS)**, **software signing**, **e-government services**, **document signing (PDF, e-filing)**, and **secure email systems** such as PGP.

In short:

An RSA digital signature authenticates the sender and protects the integrity of the message by using a **private key to sign** and a **public key to verify**, making it a secure and legally recognized method of digital authentication.

2.4 Hash Function:

A **hash function** is a mathematical algorithm that converts input data of **any size** into a **fixed-length output**, called a **hash value** or **message digest**. It is widely used in **cryptography** to ensure **data integrity** and security.

Key Points:

1. **Fixed-Length Output:**

Regardless of the size of the input message, a hash function always produces an output of fixed length. For example, **SHA-256** produces a 256-bit hash.

2. **One-Way Function:**

Hash functions are **irreversible**. It is computationally infeasible to retrieve the original data from the hash value, ensuring security.

3. **Integrity Verification:**

Even a small change in the input data results in a completely different hash value. This property helps detect **data tampering**.

4. **Common Hash Algorithms:**

Popular hash functions include **MD5**, **SHA-1**, **SHA-256**, and **SHA-3**. Modern systems prefer SHA-256 or higher for better security.

5. **Applications:**

Hash functions are used in **digital signatures**, **password storage**, **message authentication**, **file integrity checking**, and **blockchain technology**.

In short:

A hash function ensures **data integrity and security** by converting data into a unique, fixed length digest that is fast to compute and difficult to reverse.

2.5 Working of Digital Signature:

The working of a digital signature explains how an electronic document is **signed and verified** to ensure **authentication, integrity, and non-repudiation**. It mainly uses **hash functions** and **public-key cryptography**.

Steps Involved:

1. **Creation of Message Digest:**
   The sender applies a **hash function** (e.g., SHA-256) to the original message. This generates a fixed-length **hash value (message digest)** representing the content.

2. **Signing the Message:**
   The sender encrypts the hash value using their **private key**. This encrypted hash forms the **digital signature**.

3. **Transmission:**
   The sender sends the **original message along with the digital signature** to the receiver.

4. **Verification by Receiver:**
   The receiver decrypts the digital signature using the sender's **public key** to obtain the original hash.
   The receiver also applies the same hash function to the received message to generate a new hash.

5. **Validation:**
   If both hash values match, the digital signature is **valid**, confirming:
   Authentication of the sender
   Integrity of the message
   Non-repudiation (sender cannot deny signing)

In short:

A digital signature works by **hashing the message, encrypting the hash with a private key, and verifying it using the public key**, ensuring secure and trustworthy electronic communication.

2.6 Electronic Governance:

Electronic Governance (E-Governance) refers to the use of **information and communication technologies (ICT)** by the government to deliver services, exchange information, and interact with citizens, businesses, and other government departments in an **efficient, transparent, and secure** manner.

Key Points:

1. **Use of ICT in Government Services:**
   E-Governance uses computers, internet, and digital platforms to provide government services such as online applications, payments, certificates, and information access.

2. **Improved Efficiency and Transparency:**
   It reduces paperwork, delays, and corruption by automating processes and making government operations more transparent and accountable.

3. **Citizen-Centric Services:**
   E-Governance enables citizens to access services anytime and anywhere, improving convenience and participation in governance.

4. **Legal Recognition under IT Act:**
   The **Information Technology Act, 2000** provides legal validity to electronic records and digital signatures, supporting e-governance initiatives.

5. **Applications of E-Governance:**
   Includes **e-filing of taxes**, **online voting systems**, **digital land records**, **Aadhaar services**, **e-procurement**, and **online grievance redressal systems**.

In short:

Electronic Governance uses **digital technology** to deliver government services efficiently, transparently, and securely, strengthening the relationship between the government and citizens.